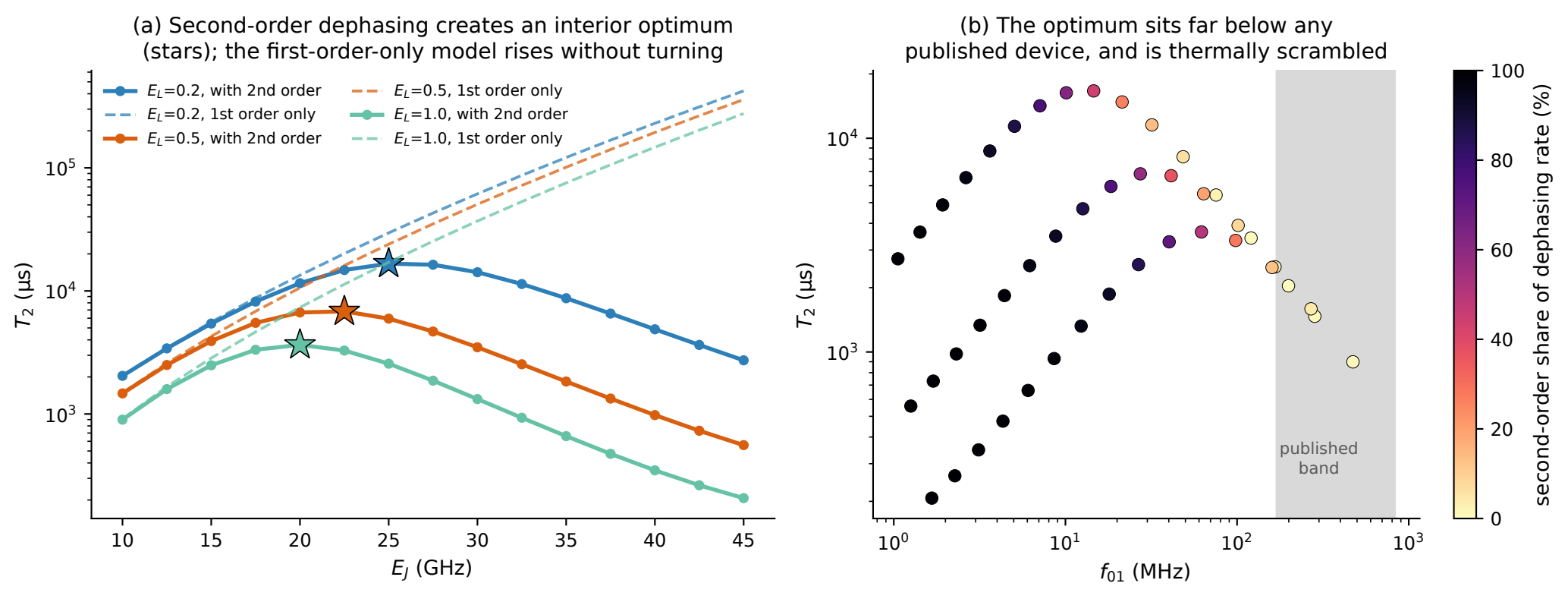

(a) Second-order dephasing creates an interior optimum (stars); the first-order-only model rises without turning
E_L=0.2, with 2nd order
E_L=0.2, 1st order only
E_L=0.5, with 2nd order
E_L=0.5, 1st order only
E_L=1.0, with 2nd order
E_L=1.0, 1st order only
T_2 (μs)
E_J (GHz)
(b) The optimum sits far below any published device, and is thermally scrambled
published band
f_{01} (MHz)
second-order share of dephasing rate (%)